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Research on the "VR+" Music Education Model and Its Practice in the Context of the Internet

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ABSTRACT

This article provides an exploration of the theoretical foundations, potential benefits, and framework of the "virtual reality plus" (VR+) music education model. This study demonstrated that by leveraging virtual reality technology, the VR+ music education model created immersive, three-dimensional environments that offered students access to diverse music learning resources. This approach offers the potential to significantly enhance students' interest in learning, as well as improved musical perception and creative abilities. The widespread adoption and implementation of this model faces challenges, however, including technical limitations, high cost, and inadequate teacher training. The future evolution of the VR+ music education model must prioritize AI-driven smart feedback systems, personalized learning paths tailored to individual skills, and interdisciplinary integration with music psychology and real-world performance simulation, to nurture musical talent by uniting creative thinking with practical expertise.

KEYWORDS

Music Education, Pattern Construction, Potential Advantages, VR+

INTRODUCTION

In recent years, the widespread reach and impact of internet technology has significantly transformed how various industries operate (Fernández-Batanero et al., 2024). In the field of education, internet technology has not only overcome the time and space limitations of traditional education but also continues to drive innovation and advancement (Deng et al., 2024; Dhakshan et al., 2024); online learning, distance education and the smart classroom are examples of emerging educational models that utilize internet technology (Singh et al., 2024). The integration of internet technology has enhanced teaching methods, while also increasing the accessibility and equity of educational resources (Ma et al., 2024).

"Virtual reality plus" (VR+) technology integrates internet technology and virtual reality (VR) technology—this synergy generates a simulated three-dimensional environment that allows users to experience a virtual world in an immersive way (Ying, 2024). These immersive, interactive experiences have shifted education in a new direction (Meng et al., 2024). In music education, VR+ has been shown to offer students rich and interactive learning experiences—such as playing virtual instruments and attending virtual concerts—while also boosting their interest, enhancing musical perception, and fostering creativity by simulating real-life music environments.

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In the context of internet-based education's rapid growth, traditional music education has struggled with limited interactivity, lack of immersive experiences, and unequal access to quality resources. This study explored how integrating VR into music education enhanced student engagement and learning outcomes, at the same time addressing a research gap regarding effectively combining VR technology with curriculum design and teaching methods in online settings.

At present, there has been limited research and application of VR+ technology in Chinese music education—the focus has been largely on the use of virtual instrument performance and music creation, as Gao and Li (2024) explored with their research into the use of VR technology with clarinet instruction. Gao and Li's (2024) findings validated the feasibility of virtual instrument design in this study, inspiring the design of realistic and user-friendly virtual instruments in the VR+ music education model.

Much existing research has focused on technical realization and preliminary application effect; research into the VR+ music education model's system construction, implementation strategy and long-term practical effects is, however, lacking (Yuldashev, 2024). Yuldashev (2024) discussed new opportunities for composers and performers operating within a virtual reality platform—this was relevant to this VR+ study, as it provided a theoretical basis for expanding creative and performance spaces in music education using VR technology. The study presented in this article incorporated concepts like those discussed in Yuldashev's work, to encourage students in exploring their creativity in virtual music composition and performance in the VR+ music education model.

The significance of this study lies in providing educators with practical strategies to leverage VR technology when using online music education, while also addressing the limitations apparent in traditional online music teaching, For example, by introducing immersive learning experiences and personalized learning paths, music education can become more engaging and effective, ultimately improving its overall quality in the digital age. Additionally, this study contributes to the theoretical understanding of how VR technology can reshape music education, offering new perspectives for curriculum innovation and teacher training in an internet-based context. It is hoped that this research will support the popularization and application of the VR+ music education model and thereby contribute to the advancement of music education.

LITERATURE REVIEW

Challenges of Traditional Music Education in the Online Era

Wang (2023) pointed out the current lack of resources and ill-structured curriculum in music education. In an online learning environment, these issues have become even more prominent. For example, traditional online music courses often rely on pre–recorded videos and static course materials, many of which lack real–time interaction and immersive experiences. This limits students' active participation and makes it difficult for students to fully understand and master music knowledge and skills. Existing research on solving these problems has been largely theoretical—few studies have proposed practical solutions, especially solutions using innovative technologies like VR. This leaves a significant gap, one that this study aimed to fill by exploring how VR-integrated music education was able to address these resource-related and interactivity-related problems.

Current Research on VR-Based Music Education and Its Methodological Limitations

Gao and Li (2024) demonstrated the feasibility of virtual instrument design in this study through their investigation of VR technology in clarinet instruction. Their study, however, mainly focused on the technical realization of virtual instruments and its short-term effects on clarinet pedagogy. In addition, the study's sample size was relatively small, involving only a limited number of clarinet students, thus the sample size may not have been representative enough to draw general conclusions regarding the impact of VR-based music education with different musical instruments and diverse student groups.

In discussing new opportunities for composers and performers in virtual reality, Yuldashev (2024) provided a theoretical basis for the expansion of creative and performance spaces in music education through VR technology, but did not explain how these opportunities might be integrated into a music education curriculum, giving teachers no clear guidance on how to design activities to make the most of these opportunities—an important element that this research will focus on.

In general, existing studies on VR–based music education have concentrated on either technical implementation, or short-term application outcomes (e.g., immediate student engagement or basic skill acquisition). Currently, there is insufficient research on system construction, implementation strategy, and the long-term practical effects of the VR+ music education model, amplifying the need for this study, with its comprehensive exploration of the VR+ music education model from multiple perspectives, including curriculum design, innovative teaching approaches, and longitudinal evaluation of teaching.

Teaching Strategy Insights From Literature on VR Music Education

In evaluating teaching strategies, Lin (2022) emphasized the integration of theory and practice in music education—inspiring this study's design of practical VR teaching activities. Previous studies, in applying such strategies, faced challenges regarding implementation, such as a lack of effective teaching tools. In the context of VR-integrated music education, this study aimed to use VR technology to overcome such implementation difficulties by creating virtual concert scenarios in which students might better apply music theory and knowledge in practice.

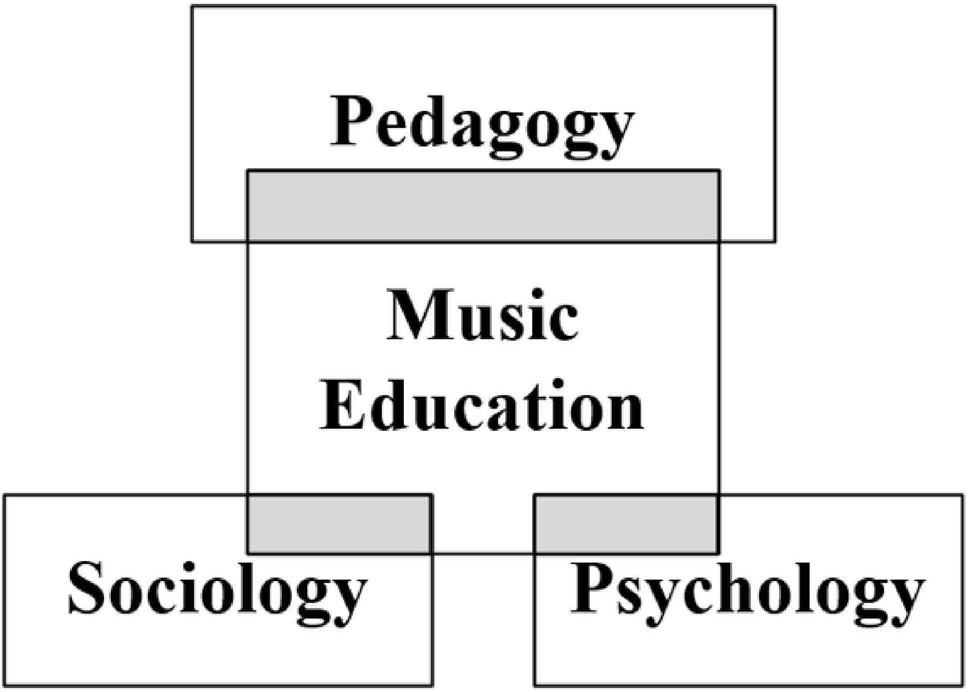
In summary, a thorough review of existing literature revealed the need for more in-depth research on how to effectively optimize the integration of VR technology into internet-based music education. This study, therefore, focused on filling research gaps by exploring a comprehensive VR+ music education model that addressed curriculum design, teaching methods, and long-term learning outcomes.

MATERIALS AND METHODS

Theoretical Basis of VR+ Music Education

Music education does not only involve the teaching of artistic skills, but integrates elements drawn from many disciplines—forming a rich and diverse knowledge system, as shown in Figure 1. Music psychology—involving students' cognitive process, emotional response and learning mechanisms—helps teachers to better understand individual learner differences and design more efficient and personalized practice strategies and teaching methods (Chen, 2024). The sociology of music broadens student horizons, exploring how music interacts with societal structures and cultural traditions; it helps to reveal the social function and cultural diversity and their roles in building social identity—this is an important part of cultivating cross-cultural understanding and social responsibility (Liu, 2024). Music pedagogy bridges theory and practice; it not only pays attention to the science of curriculum design, but studies the effectiveness of different teaching methods and the appropriateness of assessment strategies (Barrett et al., 2024). Utilizing these different disciplines enriches music education and enhances music's educational value and social influence.

Figure 1. Multidisciplinary music education



Based on these theoretical foundations, some scholars have proposed that VR+ technology can be effectively integrated into music education to give students an enhanced learning experience (Zhang, 2024). VR+ technology represents the cutting-edge trend of comprehensively integrating information technology with the help of advanced computer simulation technology, thereby carefully constructing a realistic three-dimensional virtual space. This space offers users an unprecedented sensory experience—in this virtual world, vision, hearing and even touch are skillfully integrated, making users feel as if they are in a completely immersive environment. Users not only interact with the virtual world through natural body posture and touch, but they receive accurate feedback from the virtual environment in real time—greatly enhancing the user's participatory experience. VR+ technology can both realistically simulate real-world environments and go beyond physical limitations by creating imaginative scenes that are impossible in real life, significantly expanding users' cognitive boundaries and imagination.

VR technology has catalyzed a profound change in education. By simulating historical scenes, anatomical structures, and physical experiments, this technology has significantly enhanced student interest and engagement; additionally, it has improved students' problem-solving skills. All of this offers vast new potential developments in education.

Advantages of VR+ Music Education

In the field of music education, the integrated application of VR+ technology offers many advantages, all of which have begun to affect traditional teaching modes (Safian et al., 2024). Through innovative design of virtual instruments, students have been provided with an effective way to practice, without needing to buy expensive physical instruments—students need only to wear a VR device and they can instantly be placed in a world full of infinite musical possibilities. With the touch of their fingertips, for example, they can visually and intuitively feel the piano keys and perform classical piano music, or enjoy the rhythms playable on the modern electronic organ. All musical instruments are within reach—this will undoubtedly reduce the initial cost of learning an instrument and enable more students, with no financial burden, to embark on their musical journey (Yang, 2024).

VR+ technology enables students to master various instruments by virtue of interactive experiences in a virtual environment—one without physical instruments. This learning method can broaden students' musical horizons; it also stimulates curiosity regarding different musical styles. Students can switch musical instruments freely in a virtual world; from strings to wind music, each instrument can be vividly experienced. It is worth noting that the VR system can detect student performance in real time, accurately capturing the accuracy and strength of each note, as well as rhythm and dynamic expression when playing. VR+ technology can provide students with detailed feedback and suggestions; this instant feedback mechanism is equivalent to providing each student with a private music tutor who can point out errors in a timely manner and offer targeted suggestions. Such personalized teaching improves learning efficiency and helps students develop a clear understanding of their own performance abilities, supporting the improvement of their technical skills (Kutlimuratovich, 2024). Additionally, with the advantage of placing no financial burden on students, VR has presented a bold breakthrough in enabling broader access to music education.

In addition, VR technology can simulate real performance environments, from concert halls to recording studios—even the natural sound environment of the outdoors. Students can then learn to adapt to the performance requirements of different environments; they can also perform for a simulated audience, effectively improving their confidence and expressive skill. Such immersive learning helps students feel as if they were in a real environment—each exercise is like a small live performance. These immersive experiences greatly enhance student engagement and assist their learning in practical ways.

Construction of the VR+ Music Education Model

In constructing the VR+ music education model, the first and most important factor was that it should build a bridge that enabled seamless, real-time communication and feedback between teachers and students to create dynamic, interactive learning. To build a truly scalable and implementable VR music education model it was necessary to create a simulated virtual scene and integrate it with a sensory experience, so that students could, as it were, enter into a "real" concert hall, deepening both their enthusiasm and the emotional resonance of their playing. This immersive experience was to be a bridge between students and the music world, as well as a way to stimulate creativity. In addition, it was necessary to explore the creation of innovative teaching methods, content, and assessment strategies—educational needs and technology change every day, and education can remain at the forefront only through continuous innovation (Sai, 2024).

When constructing the VR+ music education model, it was necessary to create the virtual classroom—the center of all learning activities—so that it retained all the functions of a traditional classroom, yet was enhanced through an intelligent, three-dimensional spatial layout. The virtual musical instruments needed accurate physical simulation and elaborate interactive design to enable students to easily cross into an imaginative musical world where they could enjoy learning all kinds of musical instruments, whether classical or modern, in solo or ensemble formats. The virtual concert, as an important platform for practical training, simulated the atmosphere of a real concert, allowing students to show what they had learned—students performed on a virtual stage and received feedback from a virtual audience. This improved their performance confidence and capacity to be full "in charge" on stage. This VR learning environment needed to include auxiliary elements, such as a virtual music library and a virtual composition studio—providing vast music resources and space for learning and music creation.

The VR+ music education model needed to follow a set of rigorous and systematic steps to ensure high quality education with strong results. First, teachers needed to analyze the teaching objectives and students' individual characteristics, then thoughtfully design a comprehensive lesson plan that included virtual instrument skills, music theory, music creation, and appreciation. They also had to prepare well-matched VR teaching resources to provide a strong foundation for students' learning experience. Subsequently, students wore VR equipment, to enter a virtual classroom full of fantastic colors. Here, they practiced with virtual instruments, participated in virtual concert performances, and made full use of the powerful interactive function of the VR platform to communicate with teachers and classmates in real time and share the beauty of music.

VR tasks included simulating the performance of classic musical pieces, such as Beethoven's *Moonlight Sonata*, Mozart's *Symphony No. 40*, and Chopin's *Nocturne in E-flat Major*, *Op. 9, No. 2*. Teachers integrated VR-based feedback by closely observing students' performance data in real time. For example, if a student made timing mistakes in the *Moonlight Sonata*, the teacher was able to use the VR platform's annotation function to mark the incorrect parts and provide video demonstrations of the correct rhythm. When multiple students had difficulty with a particular musical piece, the teacher adjusted the teaching plan to include more practice on that piece, or related techniques. Teachers paid attention to students' learning progress and performance through an advanced monitoring system. They could then provide timely, personalized guidance and feedback to help students progress. Teachers adjusted their instructional strategies based on the real-time data from the VR system. For example, if the VR system showed that a student was struggling with the fingering of a virtual guitar, the teacher used the VR platform's demonstration feature to show the correct fingering positions in clear steps. If most students in the class had difficulty understanding a certain music theory concept during the virtual lesson, the teacher immediately added, into the VR environment, more interactive examples and practice exercises related to that concept.

Teachers provided personalized learning plans for students with different learning needs. For students who quickly mastered the basic skills, the teacher assigned more advanced musical pieces and complex techniques for practice. For students making slower progress, the teacher focused on reinforcing basic knowledge and skills. Finally, the students' mastery of music knowledge, skill level and innovation ability were monitored and assessed using evaluation methods such as virtual tests, online homework and virtual concert performances, so as to ensure the successful achievement of learning objectives. At the same time, students were encouraged to conduct self-reflection and peer evaluation to stimulate autonomous learning and teamwork.

Having now established the framework of the VR+ music education model, the following sections will detail the experimental design, procedures, and result analysis conducted to evaluate this model's effectiveness.

The artificial intelligence virtual teaching platform created by Nanjing Normal University Conservatory of Music is an example. This platform was created jointly by Nanjing Normal University Conservatory of Music and Nanjing Normal University Computer College. In this study, the data was collected from a single class in the first grade of Nanjing Normal University Conservatory of Music. A total of thirty students participated in the study. During the data collection period, students used the VR+ music education system in groups. Usually, five students used VR equipment simultaneously under the guidance of one teacher. The teacher played a crucial role in the process. Before the VR class, the teacher was responsible for designing teaching plans, which included determining the learning objectives, selecting appropriate virtual instrument operation skills, and content regarding music theory knowledge, music creation and appreciation. During the class, the teacher monitored students' learning progress through the VR platform's advanced monitoring system and provided personalized guidance and feedback in a timely manner. Each VR class lasted approximately 45 minutes.

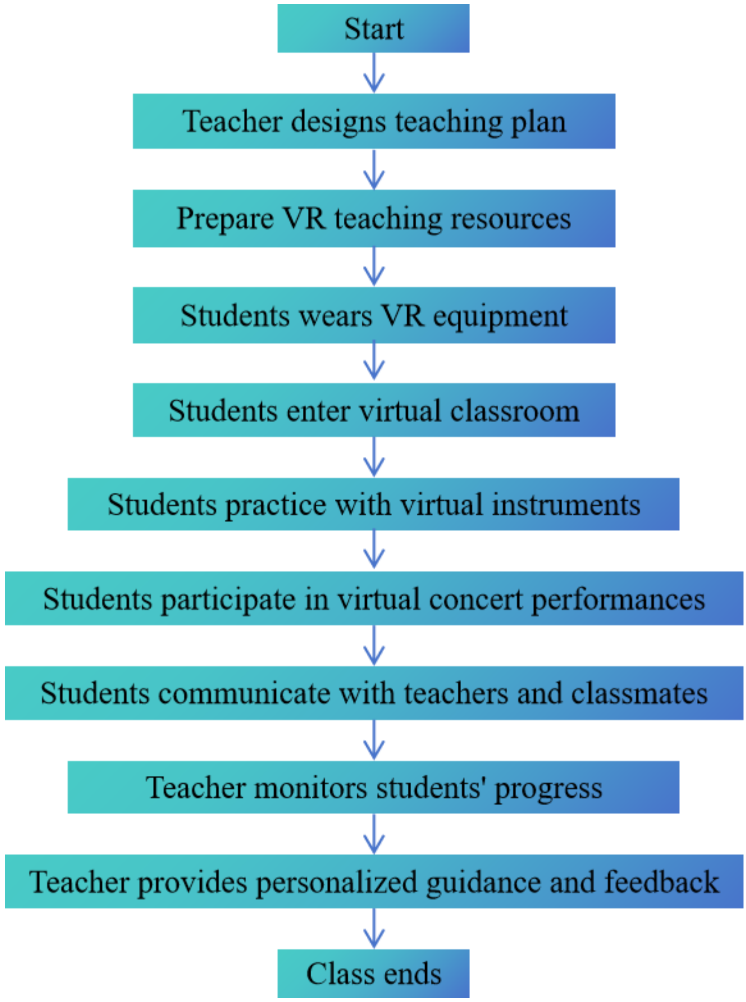
The data was collected through multiple methods. First, students filled out questionnaires before and after the VR+ music classes. The questionnaires covered aspects such as learning interest, participation, and practical ability. Data collection lasted for sixteen weeks. The student questionnaire consisted of thirty questions in total. There were ten multiple-choice questions to collect basic information such as students' prior music experience. Fifteen Likert scale questions were used to measure student attitudes towards VR+ music education, with possible responses including: [1] strongly disagree; [2] disagree; [3] neutral; [4] agree; [5] strongly agree. In addition, there were five open-ended questions for students to answer in a way that freely expressed their opinions and suggestions regarding the VR+ music classes. In addition to the questionnaires, students' virtual concert performance videos were collected as an additional data source to evaluate practical skills and musical expressiveness.

Sample items of the Likert scale questions in the student questionnaire included: "I am interested in learning music through VR technology;" and, "VR technology helps me better understand music theory." Among the Likert scale questions, five were used to measure students' learning interest, five measured their perception of the effectiveness of VR in music learning, and five provided self-assessment of learning progress. To ensure the validity and reliability of the questionnaires, several statistical analyses were conducted. For the student questionnaires, the Cronbach's α coefficient was calculated to assess internal consistency. The result showed that the Cronbach's α coefficient was 0.85, indicating good internal consistency reliability. For the construct validity, factor analysis was performed. The factor analysis results supported the construct validity of the questionnaire, with three factors extracted that corresponded well to the three main aspects of learning interest, participation, and practical ability.

Second, teachers filled out teaching feedback forms that evaluated teaching satisfaction, teaching efficiency, and students' progress speed. In total, thirty students completed the student questionnaires, and five teachers completed the teaching feedback forms. All responses were carefully collected and analyzed to obtain the results presented in this study. In addition to the basic VR technology, the platform was also equipped with various innovative and auxiliary teaching systems, such as an automatic harmony allocation system.

The experiment was conducted from September 1, 2024, to December 31, 2024. A class was randomly selected from first-year college students for investigation and the effects of VR+ music education model were analyzed. Figure 2 shows the sequence of steps in the VR+ music education experiment, from teaching plan design to the end of the class.

Figure 2. Experimental process flowchart



RESULTS AND DISCUSSION

The research adopted a mixed methods approach, combining quantitative and qualitative research methods. The population of this study included thirty first-year university studentsat Nanjing Normal University Conservatory of Music and their music teachers. The five teachers were selected because they had direct teaching experience with the students in the sampled class, enabling them to provide valuable insights into the teaching process. Given the large population size, a single class was selected as the sample. This sampling method was a form of convenience sampling; although not be fully representative of the entire population, it was chosen due to the practical constraint of limited time and resources. In addition, utilizing a single class allowed for more detailed data collection and analysis.

A quantitative approach was used for data analysis. Descriptive statistics were calculated for students' learning experience data, including their interest in learning, level of participation, and practical skills. To analyze the changes, the percentages of students' responses before and after the VR-integrated music classes were compared. Regarding teacher feedback data—including teaching satisfaction, instructional efficiency, and students' learning progress—the mean values before and after the implementation of the VR-integrated music education model were calculated and compared.

Descriptive statistics were used to assess the improvement of teaching quality, and assess the distribution of students' satisfaction levels before and after teaching. The yearly growth or decline in the proportion of different satisfaction levels was calculated to evaluate the impact of the VR+ music education model on teaching quality.

Students' Learning Experience

In the VR+ music classroom, students played various musical instruments in a virtual environment, experiencing different music activities and ways of playing through the VR head display and controller. First, students were immersed in the virtual music environment and were able to then intuitively experience the playing capability and timbre of different instruments—this enhanced their learning interest and participation. Second, VR technology provided rich opportunities for interaction. Students were able to rehearse with virtual musicians and even participate in virtual concerts. This interactive experience enabled students to understand music theory and playing skills in more depth. In addition, the VR+ music education model expanded limitations imposed by time and space, so that students could study anytime and anywhere. This flexibility greatly improved students' learning motivation and satisfaction.

To determine whether the changes in students' learning experiences were statistically significant, paired samples *t*-tests were conducted. The results showed that for learning interest, the *t*-value was 5.67 ($p < 0.01$), indicating a significant increase. For participation, the *t*-value was 4.89 ($p < 0.01$), also showing a significant increase. Regarding practice ability, the *t*-value was 6.23 ($p < 0.01$), demonstrating significant improvement.

As shown in Table 1, the proportion of students expressing strong interest in learning increased from 60% before VR+ music class, to 90% after the class; the proportion of students actively participating in the process rose from 70% to 95%. This demonstrated that the VR+ music education model effectively stimulated students' learning interest and participation.

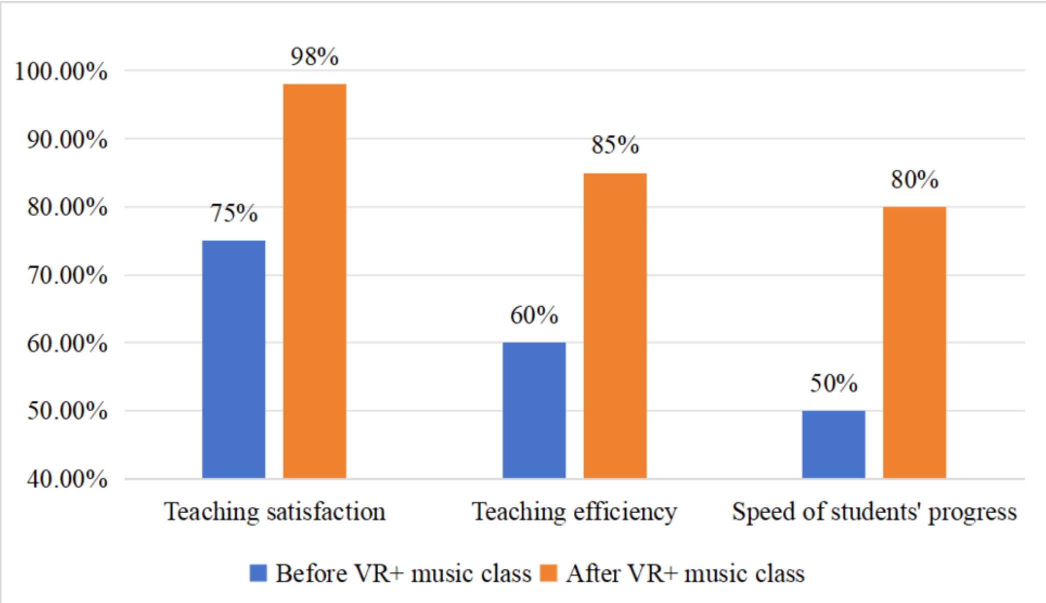
Table 1. Investigation class learning experience

Project	Before VR+ music class	After VR+ music class
Learning interest	60% Expressed strong interest	90% Expressed strong interest
Participation	70% Actively participated	95% Actively participated
Practice ability	50% Mastered basic skills	85% Skillfully played several instruments

Teacher Feedback

As shown in Figure 3, by utilizing the VR+ music education model the teaching experience improved significantly. Teachers discovered that using virtual reality technology made the classroom livelier and more interesting, and that students studied with more enthusiasm. VR technology simplified the operation and demonstration of musical instruments and made explanation easier to grasp. It also allowed teachers to create diverse teaching activities within the virtual environment, such as simulated concerts and music studios, which enriched teaching methods. In addition, the VR platform also provided a real-time feedback function that enabled teachers to recognize student progress in real time and adjust teaching strategies where needed. This not only lightened teachers' burdens, but improved teaching efficiency and quality. Overall, teachers increased their positive expectation and confidence in the VR+ music education model.

Figure 3. Teaching feedback survey



Paired sample *t*-tests were used to test the significance of the changes in teachers' pedagogical feedback. The results showed that for teaching satisfaction, the *t*-value was 4.21 ($p < 0.01$), indicating a significant increase. For teaching efficiency, the *t*-value was 3.85 ($p < 0.01$), showing a significant improvement. And for the speed of students' progress, the *t*-value was 4.56 ($p < 0.01$), demonstrating a significant increase.

From the teaching feedback in Figure 3, teaching satisfaction increased from 0.75 to 0.98, teaching efficiency increased from 0.60 to 0.85, and the speed of student progress increased from 0.50 to 0.80. These data indicate that the VR+ music education model not only enriched teaching methods, but improved teaching quality and student learning efficiency.

Improvement in Teaching Quality

A positive learning experience encourages students to engage more actively in their learning—"Interest is the best teacher." The rich teaching resources inherent in VR technology improved students' practical ability and musical expression; they also improved the overall quality of teaching in a targeted way to different levels of student ability—which enhanced students' practical learning. In addition, the personalized learning approach enabled by VR technology met the diverse learning needs of students and directly supported their individual development, thereby enhancing teaching efficiency and learning outcomes.

To assess the significance of the change in students' satisfaction levels, an independent sample *t*-test was conducted. The results showed that the *t*-value was 7.89 ($p < 0.01$), indicating a significant difference in the distribution of satisfaction levels before and after teaching.

The improvement in teaching quality was further supported by the data in Table 2. The proportion of students with high satisfaction levels (satisfied and very satisfied) increased significantly, while the proportion of students with low satisfaction levels (very dissatisfied, dissatisfied, and average) decreased substantially. This clearly shows that the VR+ music education model was valued highly by students, and effectively improved teaching quality.

Table 2. Overview of satisfaction data in the virtual reality plus-based music classroom

Satisfaction	Satisfaction before teaching	Post-teaching satisfaction	Yearly growth
Very dissatisfied	8.23%	0.56%	–7.67%
Dissatisfied	21.45%	4.32%	–17.13%
Average	34.56%	15.67%	–18.89%
Satisfied	25.34%	45.89%	+20.55%
Very satisfied	10.42%	33.56%	+23.14%

In addition to the satisfaction data, data was also collected regarding music theory test scores and performance evaluations. The average score of music theory tests increased: from 65 before the VR+ music class, to 78 after the class. For performance evaluations, specific indicators were used, such as intonation accuracy, rhythm stability, and musical expression. The average scores on these indicators were 60, 62, and 65 before the class; 75, 78, and 72 after the class. These scores demonstrated significant improvement.

Through the analysis of the data shown in Figure 4 and Figure 5, when using the VR+ music education model, the overall satisfaction of teaching improved significantly across all areas.

Figure 4 .Distribution of satisfaction levels before teaching

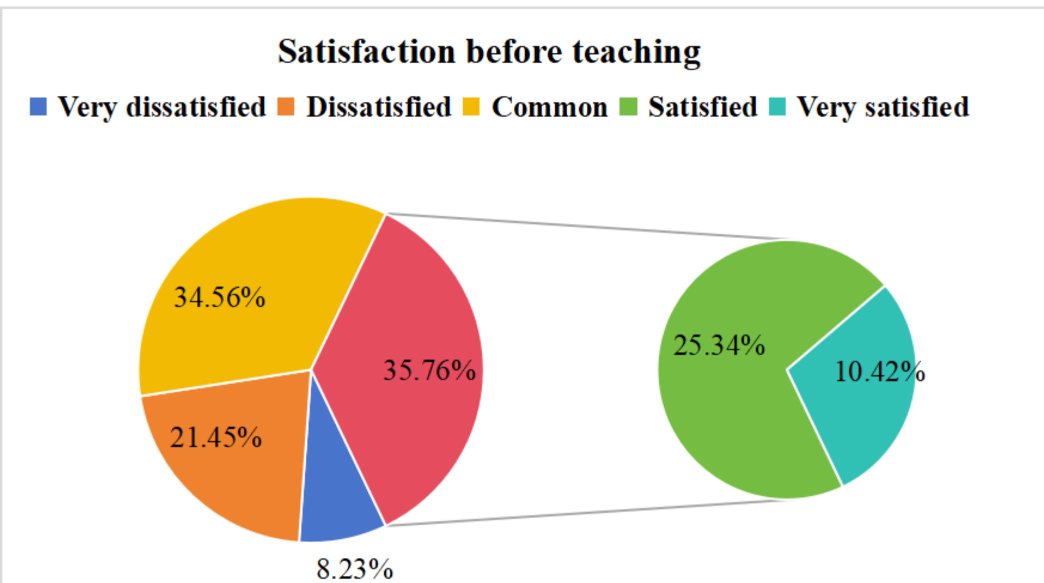
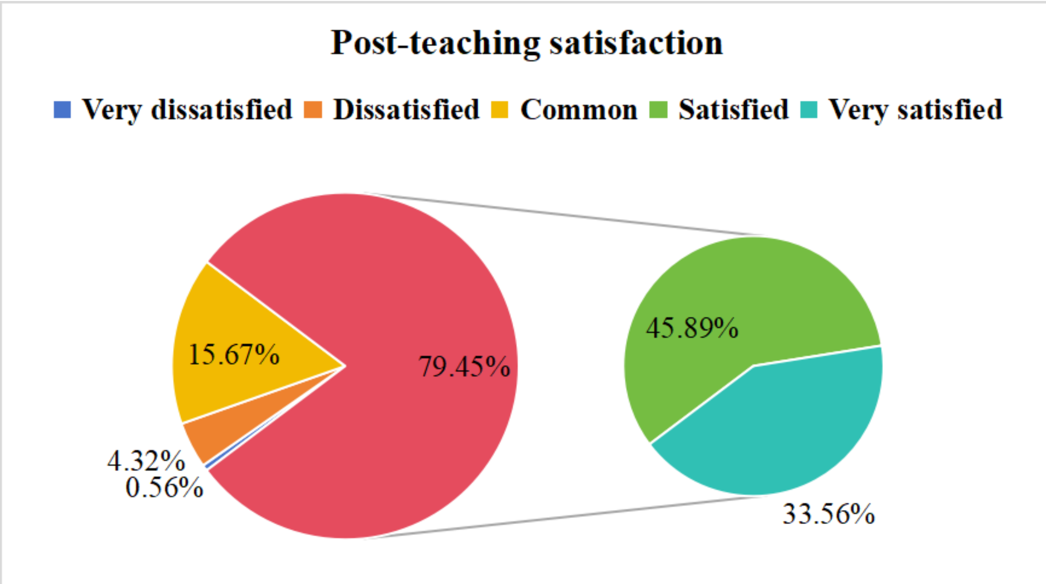


Figure 5. Distribution of satisfaction levels after teaching



To further strengthen the validity of the research findings, future studies will conduct a randomized controlled trial. In that trial, students will be randomly divided into two groups, an experimental and a control group. The experimental group will receive music education with the VR+ model, while the control group will receive traditional music teaching. Student learning achievements will be compared across music knowledge, practical skills, and musical creativity. Objective tests will be used to measure student progress, such as music theory exams and practical performance evaluations. Additionally, subjective feedback on their learning experiences will be collected, via questionnaires and interviews.

Faculty data will be compared between the two groups in terms of teaching satisfaction, workload, and the effectiveness of teaching methods. Teachers will be asked to fill out detailed teaching evaluation forms that cover classroom management, student engagement, and achievement of teaching objectives.

By comparing data from the two groups, the distinct impact of the VR-integrated music education model will be identified; additionally, that same data will provide further evidence to support its use. The interval regarding dissatisfaction (including very dissatisfied, dissatisfied and average) reached 43.69%, unequivocally demonstrating the model's significant impact on enhancing learning experience and teaching quality. Students' high recognition of the model underscores VR technology's potential in music education. These findings firmly establish the groundwork for the future research initiatives detailed below, which will systematically explore its long-term impacts and interdisciplinary applications.In promoting the widespread integration of the VR+ music education model, there will be inevitable challenges. Issues with technical resources and support seems the most prominent current problems—particularly regarding needed improvements with the immersion, interactivity and stability of VR equipment. In addition, cost is a significant factor restricting the model's popularity; the purchase, maintenance and update of VR equipment is expensive, as is the research and development of teaching content. Teachers need to master new technologies to meet the pedagogical needs of a VR environment, which increases the burden on teachers of time and cost.

Despite these many challenges, the VR+ music education model has been shown to offer unprecedented opportunities (Deliyannis et al., 2024). It expanded traditional teaching methods, making music teaching more engaging, intuitive and efficient. At the same time, through VR technology, students explored freely in a virtual environment; they enhanced their learning interest and participation and teaching quality was also improved. Students not only mastered music knowledge but exercised their technical ability and innovative thinking.

As technology improves and becomes more readily available, VR equipment will become more popular and the teaching content richer and more diverse. At the same time, the model will advance in the ways it pays attention to individual student needs and provides customized learning to help students develop their skills and expressivity (Fang, 2024).

CONCLUSION

This article has discussed the theoretical basis, potential advantages and construction of the VR+ music education model, pointing to research that demonstrates how the VR+ music education model has brought revolutionary change to traditional music education with its innovative teaching methods and rich pedagogical resources. According to the findings in this study, the VR+ music education model enhanced students' technical skills and creative thinking and demonstrated that the VR+ music education model effectively cultivated interdisciplinary talents through its innovative educational approach.

The wider application of this model faces challenges, however, such as issues surrounding technical resourcing and support, high cost, and an insufficient degree of teacher training programs. With technology continuing to advance, the VR+ music education model will increase its capacity to attend to individual learning needs and provide more targeted, efficient teaching strategies. At the same time, interdisciplinary integration is expected to become a new research trend—combining VR technology with areas such as music creation and music education psychology—to explore broader possibilities for educational applications.

It should be noted that this study had certain limitations. The sample in this study was selected from only one class of first-year university students at Nanjing Normal University Conservatory of Music. Although this sampling method was convenient for in-depth data collection and analysis with limited time and resources, it may not be fully representative of all music students. Different schools, year levels, and musical specialties may reveal different responses to the interviews and surveys. In addition, the research was conducted within one semester, a relatively short period of time and insufficient for observing any long-term impact of the VR+ music education model. Future research could expand the sample size and scope, including students from different regions, schools, and year levels, and conduct longitudinal follow-up studies to more comprehensively evaluate the effectiveness of the VR+ music education model.

COMPETING INTERESTS STATEMENT

The author of this publication declares there are no competing interests.

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